Beyond the Formal Grid: The Role of Fiscal Expenditure on Rural Unorganised Firm Performance

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Abstract

The rural unincorporated non-agricultural sector is an important source of employment and livelihoods in India, yet relatively little is known about the factors associated with enterprise performance within this highly heterogeneous segment of the economy. Using enterprise-level data from the Annual Survey of Unincorporated Sector Enterprises (ASUSE) 2023–24, this study examines the correlates of real gross value added (RGVA) among rural proprietary enterprises across 21 Indian states. The analysis combines ASUSE with district-level expenditure under the Pradhan Mantri Gram Sadak Yojana (PMGSY), allowing enterprise and owner characteristics to be studied alongside variation in the local infrastructure environment. This provides a framework for examining how internal enterprise characteristics and external local conditions jointly relate to performance among very small rural firms.

The descriptive analysis documents substantial heterogeneity across states, industries, skill groups, and firm-age cohorts. Enterprise activity is concentrated in a relatively small number of industries, while differences in fixed capital, output, ownership, and firm age indicate considerable variation in the scale and organisation of rural enterprises. Regression models relate log real GVA to enterprise size, fixed capital, firm age, technology use, owner skill and gender, household-based operation, and district-level PMGSY expenditure, while accounting for industry-by-state controls and using survey weights.

The results show that employment, fixed capital, ICT use, and owner skill are positively associated with enterprise performance, while female-owned and home-based enterprises report lower RGVA. Firm age exhibits a concave relationship with performance. District-level PMGSY expenditure is positively associated with RGVA in the full sample and in trade and other services, and its relationship with enterprise performance differs between young and older firms.

These findings highlight the importance of recognising heterogeneity within the rural unincorporated sector. Many of these firms operate at a scale far below conventional MSMEs, suggesting that policies aimed at enterprise development may need to account for differences in sector, ownership, firm age, and operating structure rather than treating rural unincorporated enterprises as a uniform group.

Keywords: Rural Enterprises; Unincorporated Sector; Fiscal Expenditure; Entrepreneurship; Firm Performance; Informal Sector; Unorganised Sector

1 Introduction

The unincorporated non-agricultural sector occupies an important place in the Indian economy. It contributes substantially to employment, output, and the broader socio-economic landscape[1] , particularly in rural areas where formal wage employment opportunities remain limited. The sector includes a wide range of activities in manufacturing, trade, and other services and provides income opportunities for low-skilled workers, household producers, account workers [2], and small entrepreneurs. In many cases, these enterprises also support local supply chains, preserve indigenous forms of production, and act as input suppliers or service providers to other sectors of the economy.

The importance of this sector is also visible in the continued policy attention given to entrepreneurship, self-employment, credit access, and small enterprise development. Over the years, the government has introduced several schemes aimed at supporting enterprise creation and expansion, including programmes such as Stand-Up India [3], Startup India [4], PM Vishwakarma [5], PM Svanidhi [6], MUDRA [7], and other Micro, Small and Medium Enterprises [8](MSME)-oriented initiatives. These programmes primarily focus on promoting entrepreneurship, facilitating access to credit,

[1] MoSPI, 2024. Press Information Bureau (2024)

[2] Own account worker is defined as one person on their own, or along with some partners, running business without hiring any paid labour.

[3] The Stand-Up India Scheme is a government program launched on April 5, 2016. It helps women and people from Scheduled Castes (SC) or Scheduled Tribes (ST) get bank loans. These loans range from ₹10 lakh to ₹1 crore to start a new business. StandUp India.

[4] Startup India aims to build a strong ecosystem for nurturing innovation, driving sustainable economic growth, and generating large-scale employment opportunities across the country. Startup India.

[5] PM Vishwakarma aims to provide holistic and end-to-end support to artisans and craftspeople through access to collateral free credit, skill training, modern tools, incentive for digital transactions and market linkage support. PM Vishwakarma.

[6] The PM SVANidhi (Pradhan Mantri Street Vendor's AtmaNirbhar Nidhi) is a micro-credit scheme launched on June 1, 2020, by the Ministry of Housing and Urban Affairs, Government of India. It provides collateral-free working capital loans to urban street vendors to help them resume and expand their businesses. PM Svanidhi.

[7] The Pradhan Mantri Mudra Yojana (PMMY) is a Government of India scheme launched on April 8, 2015. It provides collateral-free loans up to ₹20 lakh to non-corporate and non-farm micro and small enterprises.PM Mudra Yojana.

[8] As per the MSME classification (2020), enterprises are categorized as micro, small, or medium based on combined thresholds of investment in plant and machinery/equipment and annual turnover. The respective limits are ₹1 crore and ₹5 crore for micro enterprises, ₹10 crore and ₹50 crore for small enterprises, and ₹50 crore and ₹250 crore for medium enterprises.

supporting enterprise formalisation, enhancing skills and productivity, and strengthening the growth prospects of micro and small businesses. The frequent announcement and extension of such schemes also indicates that this sector remains central to India's development strategy. Yet this sustained attention itself signals persistent constraints—low capital, weak formal credit access, poor market linkages, limited technology adoption, and inadequate infrastructure—that restrict enterprises' ability to procure inputs, reach customers, or grow beyond subsistence.

Evidence from the Annual Survey of Unincorporated Sector Enterprises (ASUSE) shows that enterprise numbers, employment, and value added have all recovered post-pandemic, suggesting that enterprise[9] creation is not, in itself, a binding constraint; yet the sector remains dominated by own-account and very small firms marked by low productivity and limited digital adoption (Gautam & Dhanya, 2026). However, far less is known about the constraints that restrict the growth of informal firms and the reasons why informality persists (Raj & Sen, 2016).

A related concern is that the unincorporated sector is often discussed within the broader MSME framework. Many rural unincorporated enterprises are extremely small and household-based, and may be better understood as nano-enterprises[10] rather than conventional micro enterprises. These units may have a low fixed capital, few or no hired workers, limited access to formal credit, and weak integration with wider markets. As a result, their constraints may differ from those faced by larger micro and small enterprises that are more visible within formal MSME policy frameworks. Treating the sector as a single category can therefore lead to an incomplete understanding of how these enterprises operate.

Against this backdrop, this study addresses the following research question: what factors determine the performance of rural unincorporated enterprises in India, and to what extent does fiscal expenditure explain this performance? Drawing on enterprise level observations from ASUSE 2023–24, the analysis proceeds in two stages. It first documents descriptive patterns in enterprise activity across sectors, states, industries, and skill and age cohorts. It then examines the enterprise-, owner-, and district-level correlates of real gross value added, with particular attention to fiscal support. In doing so, the study aims to identify policy-relevant levers for nano-enterprises operating largely outside the formal gambit.

The rest of the paper is structured as follows: Section 2 reviews the related literature. Section 3 outlines the ASUSE data along with banking, fiscal and infrastructure components. Section 4 presents stylised facts from unit-level observations in the ASUSE 2023-24. Section 5 discusses the empirical methodology followed by results in Section 6. Section 7 concludes.

[9] Enterprises and firms are used interchangeably in this paper.

[10] Nano-enterprise is typically managed by a self-employed worker or informal entrepreneur engaged in very small-scale, low-capital business activities. These individuals often lack formal registration, operate in marginalized or rural contexts, and rely on household labour, limited infrastructure, and community-based resources to sustain their ventures (Ramírez-López et al., 2025).

2 Literature Review

A large body of literature has examined the determinants of firm and industrial performance, focusing on factors ranging from firm characteristics and entrepreneurial attributes to infrastructure, access to finance and institutional conditions. Given the important role of enterprises in generating output, employment and economic growth, understanding the factors that influence enterprise performance remains a key area of research. Existing studies identify several themes that are particularly relevant for explaining variations in firm performance, defined as value added, and productivity parameters in various contexts.

2.1 Firm Activity

Firm size and age are important determinants of enterprise performance. While large firms tend to be more productive, job creation and employment growth are often concentrated among young firms. Small firms account for a large share of new jobs, although much of this effect reflects their younger age profile (Acs, Armington, & Zhang, 2007; Ayyagari, Demirgüç-Kunt, & Maksimovic, 2011; Decker, Haltiwanger, Jarmin, & Miranda, 2014; Haltiwanger, Jarmin, & Miranda, 2013). However, young firms are more sensitive to business cycles than older and larger firms (Fort, Haltiwanger, Jarmin, & Miranda, 2013). Employment fluctuations among startups are procyclical, and conditions at the time of firm birth explain much of the cohort-level variation in outcomes (Pugsley & Şahin, 2019; Sedláček & Sterk, 2017). Hazard rates also decline with firm age, and larger firms tend to survive longer, partly because low-productivity firms exit early (Hopenhayn, 1992).

2.2 Infrastructure/Fiscal Support

Roads facilitate the movement of workers out of agriculture, without major changes in agricultural outcomes (Asher & Novosad, 2020). Public investment and road creation have a strong positive association with productivity in the formal sector, but no systematic association with the output of the average informal firm (Chatterjee, Lebesmuehlbacher, & Narayanan, 2021; Klapper, Laeven, & Rajan, 2006). However, PMGSY has been found to positively affect rural informal firms, with no evidence of an effect on formal enterprises (Nandwani, Roychowdhury, & Shankar, 2025). Other studies also show that new roads can support small business creation, as all-weather roads reduce seasonal disruptions and improve access to nearby markets (Agarwal, Mukherjee, & Naaraayanan, 2023; Naaraayanan, Mukherjee, Agarwal, & Ghosh, 2024; Sharma, Kant, Mishra, & Azhgaliyeva, 2022).

2.3 Unorganised Sector

The literature on the unorganised sector highlights the role of regulations, informality and firm size. Costly regulations can discourage the creation of new firms (Klapper et al., 2006). Informal enterprises avoid registration costs, taxes and regulatory requirements, which may allow them to save resources and continue operating (La Porta &

Shleifer, 2014; Williams, Martinez-Perez, & Kedir, 2017). However, the central constraint in India's unorganised sector is not firm creation alone, but firm growth. The sector is largely made up of own-account enterprises and is marked by low productivity, limited credit access and low digital adoption (Gautam & Dhanya, 2026). Informal firms are usually very small, are often managed by less educated owners and rarely transition into formal firms (La Porta & Shleifer, 2008). As countries grow and develop, the informal economy tends to shrink and the formal economy becomes more dominant (La Porta & Shleifer, 2014).

2.4 Credit Constraints

Institutional credit helps promote rural entrepreneurship and has a positive effect on firm growth (Agarwal et al., 2023; Gang, Natarajan, & Sen, 2022; Vidyarani & Mallick, 2025). Access to finance allows enterprises to invest, expand operations and manage risks. Recent evidence also suggests that electronic payments can reduce information asymmetries and improve lending opportunities for small firms (Galilea, Farazi, & Mare, 2026).

2.5 Female Entrepreneurship

Women entrepreneurs in India's rural informal sector are emerging, but their enterprises are generally smaller, more home-based and less productive. This reflects limited access to finance, markets, networks and suitable opportunities for women (Gang et al., 2022; Ghani, Kerr, & O'Connell, 2013; Naaraayanan et al., 2024).

Overall, the literature suggests that enterprise performance is shaped by both internal firm-level factors and external local conditions. Firm age, size, credit access and ownership characteristics influence productivity and survival, while infrastructure and institutional conditions affect market access and business opportunities. This provides the basis for examining enterprise performance using both enterprise-level and district-level variables.
While previous studies have examined various dimensions of enterprise performance, India's unincorporated sector has received comparatively less attention, especially in studies based on the ASUSE dataset. Given the sector's substantial contribution to employment and economic activity, this represents an important gap in the literature. The present study contributes to addressing this gap by providing empirical evidence from the ASUSE survey and linking the dataset with administrative datasets to examine how local conditions relate to enterprise performance.

3 Data

The Annual Survey of Unincorporated Sector Enterprises (ASUSE) is a nationally representative survey of unincorporated establishments engaged in manufacturing, trade, and other services (excluding construction).[11] This study uses cross-sectional

[11] The establishments that are neither registered under Companies Act, 1956 nor under Companies Act, 2013 are only covered.

data for the rural unincorporated sector from the 2023–24 round of the survey.[12] In 2023–24, ASUSE estimated approximately 7.34 crore establishments employing 12.06 crore workers across India.

To ensure consistency in merging external administrative datasets, the analysis excludes Union Territories and seven states, restricting the sample to 21 states. The study further focuses on proprietary enterprises, yielding an analytical sample representing approximately 3.52 crore establishments and 5.11 crore workers.

Data on fiscal expenditure under the Pradhan Mantri Gram Sadak Yojana (PMGSY)[13] are obtained from the official PMGSY database maintained by the Department of Rural Development, Government of India [14]. The database provides information on expenditure incurred under the rural road programme across states and over time. The study uses district-level expenditure[15] from January 2000 to March 2023.

4 Stylised Facts

This section presents a set of stylised facts on the structure of the rural unincorporated sector. These patterns describe the distribution of establishments, employment, output, and fixed capital[16] across sectors, states, industries, and firm cohorts (based on age of firm).

4.1 Overview

Employment is relatively balanced across the three sectors. However, trade accounts for the largest share of aggregate GVA despite employing a similar number of workers, suggesting higher average productivity relative to manufacturing and other services.

Table 1: Data Overview

	All-India	Manufacturing	Trade	Other Services
Count (in crore)	3.5	1.2	1.1	1.2
Workers (in crore)	5.1	1.7	1.7	1.7
GVA [17](in ₹ lakh crore)	24.5	7.1	9.8	7.6
Fixed capital [18](in ₹ thousand crore)	6.8	1.8	2.3	2.7
Turnover/Fixed capital (median)	1.4	0.7	3.6	1.3

Notes: All financial values are in real terms using GDP Deflator. Fixed capital includes the value of owned land, buildings, machinery, tools and other equipment, ICT equipment, and other fixed assets.

[12] The unit of enquiry of the ASUSE is an Establishment.

[13] Launched on 25 December 2000 to ensure all-weather connectivity to previously unconnected rural habitations, the programme emerged as a key enabler of agricultural growth, employment generation, improved access to education and health services, and poverty alleviation. (Press India Bureau, 2025.)

[14] Accessed on 3rd September 2026. (PMGSY Dashboard.)

[15] To standardise expenditure values, total credit values at district level is used. Thus, the variable of analysis becomes Cumulative PMGSY Expenditure(in₹crore)/Total Credit Value 2022-23(in ₹crore, all in real terms.

[16] Fixed Capital and Investment imply the same variable in this paper.

4.2 State Profiles

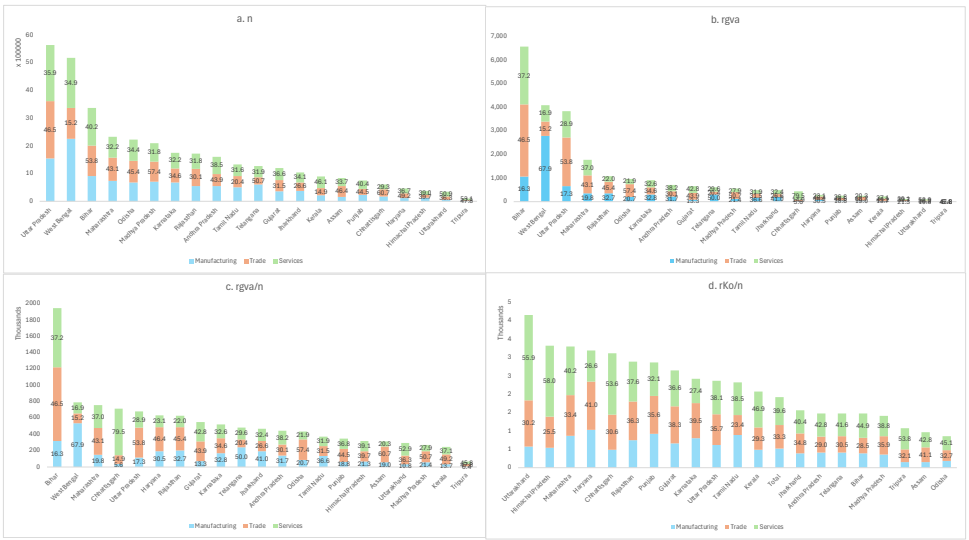


Fig. 1: State Level Profile [a. Number of Establishments, b. GVA Values (in ₹), c. GVA per establishment (in ₹), d. Investment per establishment (in ₹)]

Note: All values are in real terms.

Across the four panels, the scale of rural enterprise activity does not map neatly into stronger enterprise-level outcomes. Bihar is a notable exception, combining a large enterprise base with comparatively high aggregate GVA and GVA per establishment. Uttar Pradesh, by contrast, has a very large number of establishments but does not rank similarly high on per-establishment measures, while states such as Haryana perform relatively better on some enterprise-level measures despite a smaller overall base. Thus, the shifts in state rankings across panels suggest substantial differences in enterprise composition and scale across states.

4.3 Top Industries

A small number of industries account for a disproportionately large share of enterprise activity, with only five industries (out of a total of 68 industries) contributing over 60 percent of aggregate GVA, employment, establishments, and fixed capital, with retail trade dominating across all indicators. The concentration of activity within a small number of industries also suggests that aggregate outcomes in the rural unincorporated sector are heavily influenced by the performance of a relatively narrow set of economic activities and policies targeting a small number of industries could have broad effects.

[18] GVA has been converted into real terms using item-wise deflators, with values corresponding to the survey period from September 2023 to October 2024. Fixed Capital has been converted in real terms by deflating it with State-level GDP Deflator.

Table 2: Top 5 Industries

Industry	Sector	GVA (%)	Employment (%)	Establishments (%)	Fixed Capital (%)
Retail Trade, except of Motor Vehicles and Motorcycles	Trade	36.3	30.1	28.5	29.6
Manufacture of Wearing Apparel	Manufacturing	7.8	11.3	15.0	9.7
Land Transport	Services	15.5	8.7	11.7	12.8
Food and Beverage Service Activities	Services	3.0	6.5	4.8	4.9
Manufacture of Food Products	Manufacturing	3.2	4.7	3.7	5.4
Total share of top 5 industries		**65.8**	**61.3**	**63.8**	**62.3**

4.4 Employment Trends

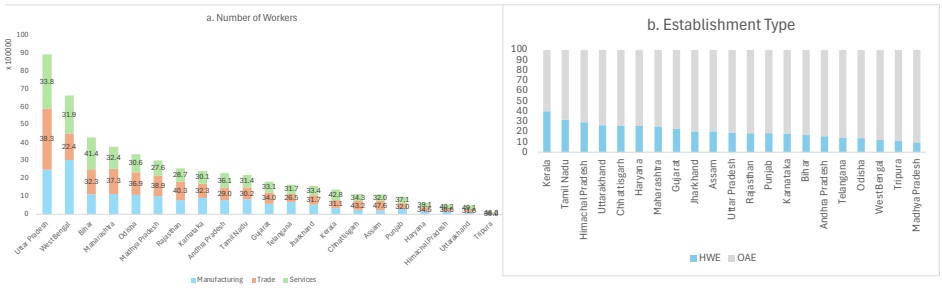


Fig. 2: Employment Trends [a. Sectorwise number of workers across states, b. Percentage of Hired Worker Establishments across States.]

Employment is concentrated in a relatively small number of states, with Uttar Pradesh, West Bengal, and Bihar accounting for particularly large numbers of workers, while the sectoral composition of employment varies across states. Despite this variation, a common feature across states is the dominance of own-account enterprises. Hired-worker establishments constitute a minority of rural unincorporated enterprises in every state, although their share is relatively higher in states such as Kerala and Tamil Nadu and considerably lower in states such as Madhya Pradesh and Tripura. Together, these patterns highlight the very small-scale and largely self-employment-based nature of rural enterprise activity.

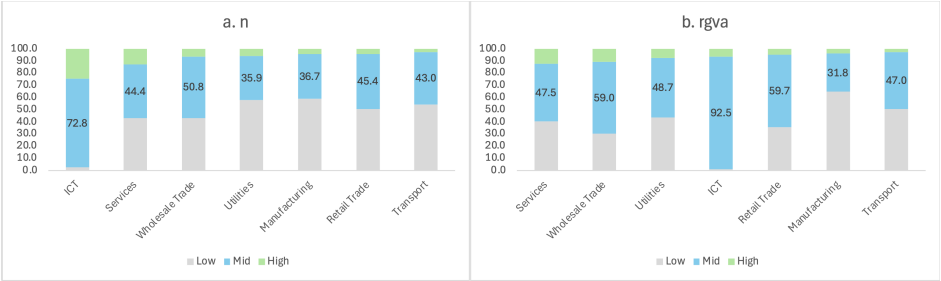


Fig. 3: Skill Distribution Within Industries [a. Share of skilled owners across industries; b. Share of skilled owners in GVA across industries]

Note: All values are in real terms.

4.5 Skill Intensity across Industries

The skill composition of rural enterprises varies sharply across industries, and the distribution of GVA does not always mirror the distribution of establishments. ICT stands out most clearly, with mid-skilled [19] owners accounting for a dominant share of both enterprises and output. In several service and trade activities, the GVA share of mid-skilled owners is also larger than their share of establishments, suggesting that enterprises run by relatively more skilled owners tend to operate at a larger scale. Manufacturing, by contrast, remains more concentrated among low-skilled owners.

4.6 Firm Age

The relatively larger presence of newer firms[20] in sectors such as ICT indicates greater recent entry into these activities, while in states such as Chhattisgarh and Tripura, younger firms account for a larger share of output relative to their share in establishments, suggesting that the contribution of new firms to output varies considerably across sectors and regions.

4.7 Year Cohorts

Cohort [21] differences are more pronounced for fixed capital than for GVA, with newer cohorts generally showing lower median capital investment while output varies within a comparatively narrower range. A notable shift appears for firms opened after 2010, when several industries display lower fixed capital among more recent entrants without a corresponding fall in median GVA. This is especially visible in ICT, where capital requirements decline sharply across newer cohorts while output remains relatively stable.

[19] For regression analysis (Section 6), mid-skill and high-skill category have been merged and is referred to as mid skill.
[20] Firm Age is calculated as: 2024 minus initial operation year
[21] Cohorts are based on initial year of operation of the firm.

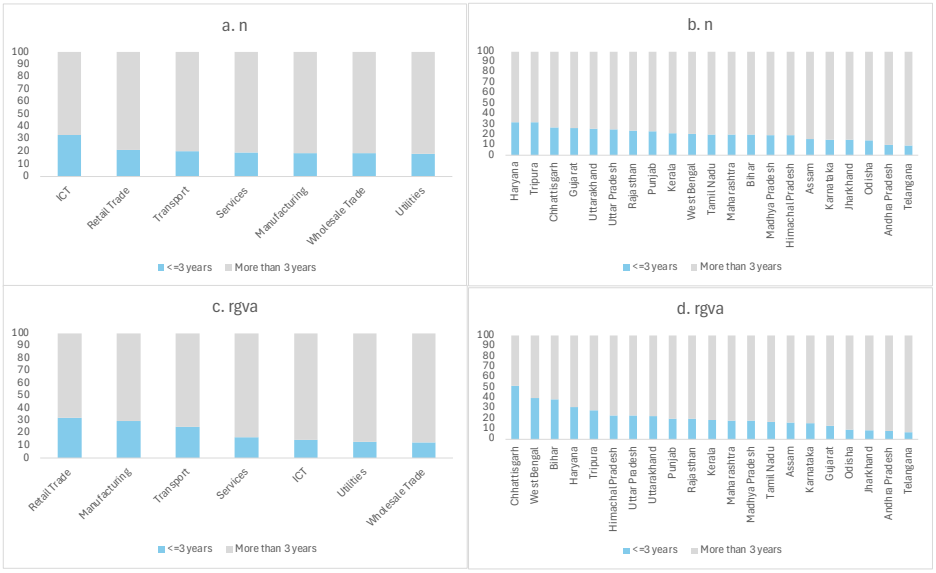


Fig. 4: Age Distribution Within States and Industries [a. Within industry share of young firms, b. Within state share of Young Firms, c. Within industry GVA share of Young Firms, d. Within state share of Young Firms]

Note: All values are in real terms.

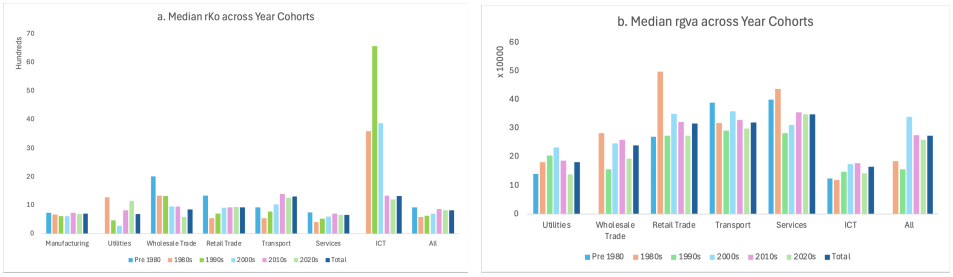


Fig. 5: : Median Value across Industries by Cohorts (in ₹) [(a) Fixed Capital, (b) rGVA]

Note: All values are in real terms.

4.8 PMGSY Trends

PMGSY expenditure varies substantially across states, with Bihar, Madhya Pradesh, Uttar Pradesh, and Odisha accounting for relatively high cumulative expenditure. In contrast, the median completion ratio is high across most states but shows less dispersion, with several states clustered close to full completion. The two panels also

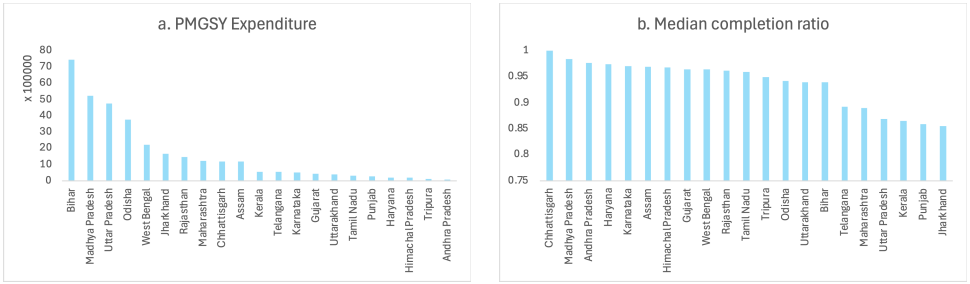


Fig. 6: PMGSY Trends [(a) PMGSY Expenditure, (b) Median Completion Ratio]

Note: All values are in real terms. Median completion ratio is the ratio between completed length of roads to adjusted sanctioned length of roads.

suggest that higher expenditure is not necessarily associated with a higher completion ratio: some high-expenditure states record only moderate completion rates, while several lower-expenditure states exhibit comparatively high completion. This points to meaningful cross-state variation not only in the scale of PMGSY spending but also in the extent to which sanctioned road works are completed.

5 Methodology

The empirical analysis examines the factors associated with enterprise performance using cross-sectional enterprise-level data from ASUSE 2023–24. Enterprise performance is measured using the natural logarithm of real gross value added (RGVA). The specifications progressively incorporate characteristics of the enterprise, the owner, and the local policy environment. Throughout, i indexes enterprises, j industries, d districts, and s states. The benchmark specification relates enterprise RGVA to measures of enterprise scale, capital, age, technology use, and location:[22]

$$\begin{aligned}\log(RGVA_{ijds}) = \beta_0 &+ \beta_1 \log(Workers_{ijds}+1) + \beta_2 \log(FixedCapital_{ijds}) \\ &+ \beta_3 Age_{ijds} + \beta_4 Age^2_{ijds} + \beta_5 Technology_{ijds} + \beta_6 WithinHH_{ijds} \\ &+ \gamma_{js} + \varepsilon_{ijds}.\end{aligned} \tag{1}$$

Here, $\log(Workers+1)$ captures enterprise employment size, with one added prior to the logarithmic transformation, while $\log(FixedCapital)$ measures the real value of the enterprise's fixed assets. Age denotes the number of years since the enterprise began operations, and the squared age term allows the association between enterprise age and RGVA to be nonlinear. $Technology$ is an indicator for whether the enterprise uses a computer or the internet, while $WithinHH$ indicates whether the enterprise operates from within household premises. The term γ_{js} denotes industry-by-state controls. These absorb factors common to enterprises operating within the same two-digit industry and state, including differences in industrial composition, market conditions, and state-specific economic environments.

[22] For all the estimations, survey weights have been used.

The second specification extends the benchmark model by introducing characteristics of the enterprise owner:

$$\begin{aligned}\log(RGVA_{ijds}) = \beta_0 &+ \beta_1 \log(Workers_{ijds} + 1) + \beta_2 \log(FixedCapital_{ijds}) \\ &+ \beta_3 Age_{ijds} + \beta_4 Age^2_{ijds} + \beta_5 Technology_{ijds} + \beta_6 MidSkill_{ijds} \\ &+ \beta_7 WithinHH_{ijds} + \beta_8 Female_{ijds} + \beta_9 (Female_{ijds} \times WithinHH_{ijds}) \\ &+ \gamma_{js} + \varepsilon_{ijds}.\end{aligned} \tag{2}$$

MidSkill identifies owners classified as mid-skilled, with low-skilled owners forming the omitted category, while $Female$ identifies female-owned enterprises. The interaction between female ownership and household-based operation is included to examine whether the association between operating from within the household differs for female-owned enterprises. This specification therefore allows owner characteristics to be considered alongside the enterprise characteristics included in the benchmark model.

The final specification introduces district-level PMGSY expenditure and allows its association with enterprise performance to differ according to the age of the enterprise. Because this specification focuses on differences between young and established firms, the continuous age and age-squared terms are replaced by an indicator for young enterprises:

$$\begin{aligned}\log(RGVA_{ijds}) = \beta_0 &+ \beta_1 \log(Workers_{ijds} + 1) + \beta_2 \log(FixedCapital_{ijds}) \\ &+ \beta_3 Technology_{ijds} + \beta_4 MidSkill_{ijds} + \beta_5 WithinHH_{ijds} + \beta_6 Female_{ijds} \\ &+ \beta_7 PMGSY_d + \beta_8 Young_{ijds} \\ &+ \beta_9 (Young_{ijds} \times PMGSY_d) + \gamma_{js} + \varepsilon_{ijds}.\end{aligned} \tag{3}$$

$PMGSY$ denotes district-level PMGSY expenditure, while $Young$ equals one for enterprises that have been operating for three years or less and zero otherwise. The interaction term allows the relationship between PMGSY expenditure and enterprise performance to vary between young and older enterprises.

All specifications are estimated using the ASUSE survey weights so that the estimates reflect the survey design. The models are estimated for the full sample and separately for manufacturing and trade and other services in order to examine whether the observed relationships vary across broad sectors. In the current specification, standard errors are clustered at the industry level, while industry-by-state controls are absorbed in all regressions.

Because the dependent variable is expressed in logarithms, coefficients on logged continuous variables can be interpreted as elasticities. For indicator variables, coefficients represent approximate percentage differences in RGVA relative to the relevant omitted category. The estimated relationships are interpreted as conditional associations

rather than causal effects, since the cross-sectional design does not eliminate the possibility that unobserved enterprise- or district-level characteristics are correlated with both the explanatory variables and enterprise performance.

6 Regression Results

This section presents report the estimates from the three specifications described in Section 5. Each model is estimated for the full sample and separately for manufacturing and trade and other services. The discussion first highlights how the estimates evolve across specifications before examining the main results by enterprise, owner, and fiscal characteristics.

Table 3: Regression Results for Model 1

Dependent variable: Log of (Real) Gross Value Added			
Independent Variables	(1) All	(2) Manufacturing	(3) Trade & Other Services
Employment	0.567***	0.890**	0.388***
	(0.106)	(0.265)	(0.087)
Log fixed capital	0.112***	0.124**	0.109***
	(0.022)	(0.045)	(0.019)
Firm age	0.020***	0.028***	0.016**
	(0.004)	(0.004)	(0.007)
Firm age squared	-0.001***	-0.001***	-0.001**
	(0.000)	(0.000)	(0.000)
Technology Use	0.251***	0.295***	0.248***
	(0.039)	(0.056)	(0.049)
Within-household enterprise	-0.287***	-0.334***	-0.249**
	(0.078)	(0.086)	(0.099)
Constant	11.056***	10.592***	11.294***
	(0.094)	(0.192)	(0.117)
Industry Controls	Yes	Yes	Yes
State Controls	Yes	Yes	Yes
Observations	35,093,260	11,656,310	23,436,950
R-squared	0.2619	0.3002	0.2284
Within R-squared	0.0434	0.0674	0.0347

Standard errors clustered at the two-digit NIC industry level are reported in parentheses. Survey weights are used in all specifications.

[1]The reference categories are no computer/internet use and enterprise located outside the household.

[2]*, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively.

The benchmark results (Table 3) show consistent positive associations of employment, fixed capital, and technology use with enterprise GVA across sectors. Firm age exhibits a concave relationship with GVA, while household-based enterprises perform relatively poorly.

Introducing owner characteristics (Table 4) leaves the coefficients on the main enterprise-level variables broadly stable. At the same time, the estimates reveal

Table 4: Regression Results for Model 2

Dependent variable: Log of (Real) Gross Value Added			
Independent Variables	(1) All	(2) Manufacturing	(3) Trade & Other Services
Employment	0.521***	0.806***	0.371***
	(0.100)	(0.254)	(0.084)
Log fixed capital	0.096***	0.106**	0.094***
	(0.021)	(0.045)	(0.018)
Firm age	0.016***	0.021***	0.014*
	(0.004)	(0.005)	(0.007)
Firm age squared	-0.000***	-0.001***	-0.000**
	(0.000)	(0.000)	(0.000)
Technology Use	0.216***	0.264***	0.209***
	(0.042)	(0.052)	(0.052)
Mid Skilled	0.099***	0.033	0.133**
(Base: Low skilled)	(0.032)	(0.039)	(0.050)
Within-household enterprise	-0.119	-0.083	-0.123
	(0.071)	(0.085)	(0.088)
Female owner	-0.254***	-0.325***	-0.224**
(Base: Male owner)	(0.082)	(0.097)	(0.084)
Female × Within-household	-0.343***	-0.298***	-0.325***
	(0.063)	(0.076)	(0.087)
Constant	11.273***	11.020***	11.377***
	(0.086)	(0.159)	(0.112)
Industry Controls	Yes	Yes	Yes
State Controls	Yes	Yes	Yes
Observations	35,093,260	11,656,310	23,436,950
R-squared	0.2731	0.3143	0.2376
Within R-squared	0.0580	0.0862	0.0462

Standard errors clustered at the two-digit NIC industry level are reported in parentheses. Survey weights are used in all specifications.

[1]The reference categories are no computer/internet use and enterprise located outside the household.

[2]*, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively.

important differences by owner skill and gender, with particularly large disadvantages associated with female ownership. The final specification (Table 5) examines whether enterprise performance is also related to district-level PMGSY expenditure and whether this association differs for younger firms.

The results across the three specifications are discussed below under three broad dimensions: enterprise characteristics, owner characteristics, and fiscal expenditure.

6.1 Firm Characteristics

The results across the three models consistently indicate that enterprise scale and capital are positively associated with real GVA. The coefficient on employment is positive and statistically significant across all specifications, with a considerably larger association in manufacturing than in trade and other services. Fixed capital is similarly positive and significant throughout, although the differences across sectors are smaller.

Table 5: Regression Results for Model 3

Dependent variable: Log of (Real) Gross Value Added			
Independent Variables	(1) All	(2) Manufacturing	(3) Trade & Other Services
Employment	0.528***	0.838***	0.356***
	(0.108)	(0.261)	(0.088)
Log fixed capital	0.099***	0.109**	0.097***
	(0.021)	(0.046)	(0.017)
Technology Use	0.234***	0.297***	0.225***
	(0.042)	(0.059)	(0.053)
Mid Skilled	0.095***	0.026	0.133**
(Base: Low skilled)	(0.032)	(0.045)	(0.052)
Within-household enterprise	-0.153*	-0.115	-0.150
	(0.081)	(0.093)	(0.101)
Female owner	-0.518***	-0.599***	-0.439***
(Base: Male owner)	(0.045)	(0.062)	(0.056)
PMGSY Expenditure	0.058***	0.073	0.054***
	(0.018)	(0.048)	(0.014)
Young firm ($\leq$ 3 years)	-0.118**	-0.070	-0.141**
(Base: $>$ 3 years)	(0.042)	(0.057)	(0.055)
Young firm $\times$ PMGSY Expenditure	0.042**	0.013	0.054**
	(0.019)	(0.046)	(0.021)
Constant	11.287***	11.047***	11.398***
	(0.077)	(0.212)	(0.093)
Industry Controls	Yes	Yes	Yes
State Controls	Yes	Yes	Yes
Observations	32,359,112	10,789,574	21,569,538
R-squared	0.2791	0.3245	0.2428
Within R-squared	0.0602	0.0882	0.0494

Standard errors clustered at the two-digit NIC industry level are reported in parentheses. Survey weights are used in all specifications.

[1]The reference categories are no computer/internet use and enterprise located outside the household.

[2]*, **, and *** denote significance at the 10, 5, and 1 percent levels, respectively.

In the benchmark specification, firm age exhibits an inverted-U relationship with GVA, suggesting that enterprise performance initially improves with age but at a diminishing rate. Technology use is also positively and significantly associated with GVA across sectors and remains stable as additional owner and fiscal characteristics are introduced. Enterprises operating within household premises, by contrast, tend to report lower GVA, although this association becomes smaller and less precisely estimated once owner characteristics are included.

6.2 Owner Characteristics

Enterprises operated by mid-skilled owners report higher GVA than those operated by low-skilled owners in the full sample, although this relationship is concentrated in trade and other services and is not statistically significant in manufacturing.

A particularly strong pattern emerges for gender. Female-owned enterprises report significantly lower GVA than male-owned enterprises across all three samples, with the largest difference observed in manufacturing. The interaction between female ownership and operation from within the household is also negative and statistically significant throughout. This suggests that the lower performance associated with female ownership is particularly pronounced among home-based enterprises, consistent with the possibility that gender-related constraints and restrictions associated with household-based production reinforce one another.

6.3 Fiscal Expenditure

Table 5 introduces district-level PMGSY expenditure and examines whether its association with enterprise performance differs for young firms. PMGSY expenditure is positively associated with GVA in the full sample and in trade and other services, while the coefficient for manufacturing is positive but not statistically significant. More importantly, the interaction between PMGSY expenditure and young-firm status is positive and statistically significant in the full sample and in trade and other services. This indicates that the association between PMGSY expenditure and GVA is stronger for enterprises that are three years old or younger than for older enterprises.
This heterogeneity is particularly relevant given the descriptive evidence on firm age and entry. While young firms report lower GVA conditional on the other characteristics included in the model, the positive interaction suggests that they may benefit more strongly from improvements in rural road infrastructure. Better connectivity may be especially important during the early stages of enterprise development, when firms are establishing access to suppliers, customers, and markets. The absence of a statistically significant interaction in manufacturing also suggests that this relationship is not uniform across sectors and appears to be driven primarily by trade and other services.
Overall, the results point to a consistent hierarchy of factors associated with rural enterprise performance. Enterprise scale, capital, and technology remain important across specifications, while owner characteristics reveal substantial heterogeneity by skill and gender. The PMGSY results further suggest that the local infrastructure environment is related not only to enterprise performance overall, but also to differences across stages of the enterprise life cycle, with younger firms appearing more responsive to rural road expenditure.

7 Conclusion

This study examines the factors determining the performance of enterprises in the Indian rural unincorporated sector. It highlights the role of enterprise and ownership characteristics such as firm age, education of the owner, technology adoption and the external environment, particularly support, in shaping outcomes.
The rural unincorporated sector is marked by substantial heterogeneity across sectors, industries, states, entrepreneurial skills, and enterprise age cohorts. Descriptive patterns indicate that enterprise activity is concentrated in a small number of industries,

while sectoral differences in GVA, fixed capital, and skill composition point to important variation in the scale and nature of rural enterprise activity.

The regression results show that enterprise performance is systematically associated with both enterprise and owner characteristics. Employment, fixed capital, and ICT use are positively associated with RGVA across specifications, while firm age exhibits a concave relationship with performance. Owner characteristics also matter: enterprises operated by more skilled owners tend to report higher RGVA, particularly in trade and other services, whereas female-owned enterprises perform worse conditional on observable enterprise characteristics. The negative interaction between female ownership and household-based operation further suggests that these disadvantages may be particularly pronounced among female-owned home-based enterprises. At the district level, PMGSY expenditure is positively associated with enterprise performance in the full sample and in trade and other services, while its relationship with RGVA differs between young and older firms.

These findings reinforce the importance of treating the rural unincorporated sector as a highly heterogeneous set of enterprises rather than as a uniform extension of the broader MSME sector. Many of these firms operate at very small scale, often with limited capital, few workers, and household-based production, and may therefore face constraints that differ substantially from those of larger formal enterprises. Policy interventions aimed at this segment may consequently need to account for differences in firm age, sector, ownership, and operating structure. The results also suggest that improvements in local connectivity and infrastructure may interact with these characteristics rather than benefiting all enterprises uniformly, highlighting the value of more targeted approaches to rural enterprise development.

The analysis is limited by its cross-sectional design, and the estimated relationships should therefore be interpreted as conditional associations rather than causal effects. Future research could exploit variation across multiple rounds of ASUSE or combine enterprise data with more detailed spatial and administrative information to examine how changes in infrastructure and local economic conditions affect enterprise outcomes over time. Extending the analysis to urban enterprises would also allow a comparison of whether the constraints documented here are specific to the rural unincorporated sector or reflect broader features of informal enterprise activity in India.

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